Sensitivity and Specificity of On-Field Visible Signs of Concussion in the National Football League

Robert J. Elbin, PhD[*]
Scott L. Zuckerman, MD, MPH[‡§]
Allen K. Sills, MD[‡§¶]
Jeff R. Crandall, PhD[||]
David J. Lessley, PhD[||]
Gary S. Solomon, PhD[‡§¶#**]

[*]Department of Health Human Performance and Recreation/Office for Sport Concussion Research, University of Arkansas, Fayetteville, Arkansas; [‡]Vanderbilt Sports Concussion Center, Vanderbilt University Medical Center, Nashville, Tennessee; [§]Department of Neurological Surgery, Vanderbilt University Medical Center, Nashville, Tennessee; [¶]Player Health and Safety Department, National Football League, New York, New York; [||]Biomechanics Consulting and Research LLC, Charlottesville, Virginia; [#]Department of Orthopaedic Surgery and Rehabilitation, Vanderbilt University Medical Center, Nashville, Tennessee; [**]Department of Psychiatry and Behavioral Sciences, Vanderbilt University Medical Center, Nashville, Tennessee

Correspondence:
Scott L. Zuckerman, MD, MPH, Department of Neurological Surgery, Vanderbilt University Medical Center, Medical Center North T-4224, Nashville, TN 37212.
Email: scott.zuckerman@vumc.org

Received, August 16, 2019.
Accepted, January 19, 2020.

Copyright © 2020 by the Congress of Neurological Surgeons

BACKGROUND: On-field visible signs (VS) are used to help identify sport-related concussion (SRC) in the National Football League (NFL). However, the predictive utility of a VS checklist for SRC is unknown.

OBJECTIVE: To report the frequency, sensitivity, specificity, and predictive value of VS in a cohort of NFL athletes.

METHODS: On-field VS ratings from 2 experts who independently reviewed video footage of a cohort of 251 injury plays that resulted in an SRC diagnosis (n = 211) and no diagnosis (n = 40) from the 2017 NFL season were examined. The frequency, sensitivity, specificity, and a receiver operating characteristic (ROC) curve with area under the curve (AUC) were calculated for each VS.

RESULTS: Slow to get up (65.9%) and motor incoordination (28.4%) were the most frequent VS in concussed athletes, and slow to get up (60.0%) was the most common VS among nonconcussed athletes. The most sensitive VS was slow to get up (66%); the most specific signs in concussed NFL athletes were blank/vacant look and impact seizure (both 100%). Approximately 26% of concussed NFL players did not exhibit a VS, and the overall sensitivity and specificity for the VS checklist to detect SRC were 73% and 65%, respectively. The VS checklist demonstrated "poor" ability to discriminate between SRC and non-SRC groups (AUC = 0.66).

CONCLUSION: In the NFL, the diagnosis of concussion cannot be made from on-field VS alone. The VS checklist is one part of the comprehensive sideline/acute evaluation of concussion, and the diagnosis remains a multimodal clinical decision.

KEY WORDS: Sport-related concussion, Football, National Football League, Visible signs, Sensitivity and specificity, Mild traumatic brain injury, American football, Concussion, Sensitivity, Specificity

Neurosurgery 0:1–8, 2020 DOI:10.1093/neuros/nyaa072 www.neurosurgery-online.com

Accurate injury identification is the critical first step in management of a sport-related concussion (SRC). Appropriate SRC identification enables prompt removal from play, which has been shown to shorten recovery.[1,2] The 3 available options for sports medicine professionals to detect acute SRC involve documentation of: (1) an athlete's symptom disclosure, (2) overt visible signs (VS) after SRC, and (3) the clinical examination. Reliance on subjective symptom disclosure may lead to missed injuries because of an athlete's poor SRC knowledge or purposeful withholding of symptoms in an effort to avoid letting teammates down or to pursue preexisting incentives.[3-6] The use of live video to document VS of SRC has been widely adopted at professional levels of play and is gaining momentum at younger levels.

The VS of SRC, as defined by the Australian Football League (AFL),[7,8] include the following: slow to get up, motor incoordination, loss of responsiveness, no protective action, clutching at head/face, blank/vacant look, impact/seizure, and/or facial injury, in addition to recording no behavior seen.[9] These VS checklists are currently being used in various forms by the National

ABBREVIATIONS: AFL, Australian Football League; **AUC,** area under the curve; **NFL,** National Football League; **NFLPA,** National Football League Players Association; **NHL,** National Hockey League; **NPV,** negative predictive value; **NRL,** National Rugby League; **OR,** odds ratio; **PPV,** positive predictive value; **ROC,** receiver operating characteristic; **SRC,** sport-related concussion; **UNC,** unaffiliated neurotrauma consultant; **VS,** visible sign

Downloaded from https://academic.oup.com/neurosurgery/advance-article-abstract/doi/10.1093/neuros/nyaa072/5820431 by Ohio State University - OARDC user on 20 May 2020

Hockey League (NHL),[10,11] AFL,[7,8] National Rugby League (NRL),[12-14] and the National Football League (NFL). Although the majority of research on the VS of SRC has examined the reliability of raters who document these VS, more recent studies have shifted focus to the sensitivity, specificity, and predictive value of the VS for predicting SRC in NHL,[10,11] AFL,[7] and NRL players.[12,13] In a study of NHL players, Echemendia and colleagues[10] reported that that loss of responsiveness, motor incoordination, and blank/vacant look were positively associated with SRC diagnoses; however, the majority of SRC (53%) had no VS. Studies from the NHL and NRL indicate that when multiple VS are considered concurrently, specificity increases.[10,12] In AFL players, Makdissi et al[7] reported that blank/vacant look and motor incoordination had high positive predictive value (PPV), and Gardner and colleagues[12] further confirmed that blank/vacant look was significantly associated with SRC diagnosis. Although the NHL, NRL, and AFL have reported similar findings across the VS of SRC, it remains unknown if these findings can be extrapolated to American professional football.

Professional American football played at the NFL level involves several unique aspects of play compared to the aforementioned sports. In the NFL, the helmet and facemask obscure the face and can complicate the assessment of blank/vacant look. Routine helmet adjustments after a collision may be mistaken for clutching at head/face. Whereas NHL, AFL, and NRL games involve collisions in proximity to the ball or puck, NFL plays involve blocking/tackling dispersed throughout the field. Lessley et al[15] examined several mechanisms of 323 NFL concussions from 2015 to 2017, and 29% occurred during blocking activities across a range of player positions and locations on the field. Furthermore, some VS, such as impact seizure, occur seconds after a collision and can be easily obscured in a congested pileup.

Given these unique mechanistic differences of American football, documenting the sensitivity and specificity of the VS of SRC in NFL players is warranted. Using a previously validated checklist of VS in SRC,[7,8] the purpose of the current study was to report the frequency, sensitivity, specificity, and predictive value of VS in a cohort of NFL athletes.

METHODS

Research Design and Procedure

A prospective, observational, cross-sectional research design was conducted. The current study was approved by the NFL, National Football League Players Association (NFLPA), and a university-affiliated Institutional Review Board (0006018120). In accordance with the NFL and NFLPA Collective Bargaining Agreement, all athletes consented a priori to deidentified injury data use for NFL Health and Safety Initiative research purposes. All reviewed videos included players who were removed from play because of concern for SRC and subsequently evaluated by the team physician and unaffiliated neurotrauma consultant (UNC).[16] Upon examination, players were then dichotomized into those who were diagnosed with an SRC and those who were not.

Participants

VS data for SRC were recorded by 2 independent expert raters for data from the 2017 NFL season.[17] Expert raters were defined as advanced degree biomechanical engineers with significant concussion and head impact research experience. These raters reviewed video footage of all injury plays and recorded VS of SRC per injury using the original AFL VS checklist (see Table 1). The reliability of these expert raters for identifying VS of SRC from video footage in NFL players has been reported.[17]

Measures and Instrumentation

Definition of Potentially Injured Players

An "injury play" was defined as a single play with concern that an SRC occurred. This most often resulted in immediate removal from play, but in the case of delayed symptom onset, medical assessment may have occurred in a locker room setting.[15] After removal from play and upon examination by the team physician and UNC,[16] players were diagnosed as having sustained an SRC or not. All concussions were evaluated in accordance with the NFL Game Day Concussion Diagnosis and Management Protocol,[16,18] and injury details were entered into the NFL's league-wide electronic medical record system. Injury data, all based on electronic health record information input solely by NFL Club head physicians and athletic trainers, were extracted by an independent injury surveillance and data analytics company. The extracted injury data, in tandem with video footage of the suspected injury play, were provided to the expert raters. Additional video footage was gathered from the NFL Injury Video Review System, and injury data related to SRC diagnosis were obtained from sideline medical evaluations conducted by game day healthcare professionals.[15]

Video Footage

NFL Films provided footage from 4 sources for injury plays. These sources included the following: (1) wide-angle end zone and sideline views; (2) network broadcast coverage, including all aired game footage; (3) untelevised NFL Films footage, often used for documentaries; and (4) additional network footage not used for televised games. All video footage was edited with in-depth video-editing software (Adobe Premiere Pro; Adobe Systems Incorporated), which allowed for zooming and viewing at slowed speeds.[15] Although each injury play had extensive video, not every case had video from all 4 sources.

SRC VS Checklist and Definitions

Raters were given a prespecified checklist of VS definitions (see Table 1), which was previously validated in an AFL study.[19] Definitions of each VS were described in detail for standardization purposes, and the presence or absence of each sign was documented by both raters. If needed, raters documented one or more VS per injury. Each video consisted of the gameplay immediately before the event, up to one minute after the event, and slow-motion replay with multiple camera angles. Raters could review the video independently as many times as needed.

Statistical Analysis

Describing the Frequency, Sensitivity, and Specificity of the VS Checklist

Descriptive statistics were used to summarize the frequency of each VS recorded by each expert rater. These VS data were then combined with NFL injury data to determine the sensitivity and specificity of

Downloaded from https://academic.oup.com/neurosurgery/advance-article-abstract/doi/10.1093/neuros/nyaa072/5820431 by Ohio State University - OARDC user on 20 May 2020

TABLE 1. Operational Definitions of VS of Concussion Taken From the AFL Checklist

Visible sign	Definition
No behavior	No observable sign witnessed
Slow to get up	Sits or lies on ground despite play continuing; not held down by another player
Motor incoordination	Appears clumsy; unsteady on feet, walks in staggered fashion, or "rubbery legs"; unable to pass or receive a ball in a coordinated fashion
Loss of responsiveness	Lying on ground for more than 1 s; does not appear to move or react or reply to game situation or others
No protective action	No effort to protect himself when falling to the ground; includes "rag doll" appearance
Clutching at head/face	Reaches for head/face with one or both hands; maintains hands on head for more than 1 s[a]
Blank/vacant look	No facial expression or emotion in response to environment or injury
Impact seizure	Tonic posturing, abnormal sustained muscle contraction; involves one or both arms; limb is held stiff despite the influence of gravity
Facial injury	Facial laceration, bleeding, or eye injury

[a]For the current study, the 1-s requirement was not utilized due to the difficulty encountered in applying this criterion to the game of American football.

VS for predicting SRC diagnosis. True positives (VS seen with positive SRC), false negatives (VS not seen with positive SRC), false positives (VS seen with negative SRC), and true negatives (VS not seen with negative SRC) were delineated. Sensitivity and specificity were then calculated with a 2 × 2 contingency table, along with PPV, negative predictive value (NPV), and positive/negative likelihood ratios for each VS. The combined average sensitivity and specificity for the VS checklist to diagnose SRC for each rater was calculated (ie, exhibiting at least 1 VS of SRC, excluding no behavior).

Documenting the Contribution of Each VS for Predicting SRC

Each rater evaluated the same injury plays, and because 2 raters were used, the total injury plays analyzed was 502. A univariate and multivariate logistic regression was performed with each VS (yes/no) as a predictor variable and concussion (yes/no) as the outcome variable for each rater. Based on these values, a receiver operating characteristic (ROC) curve was generated with a corresponding area under the curve (AUC). The AUC values were classified according to previous studies: 0.91-1 = excellent; 0.81-0.90 = good; 0.71-0.80 = fair 0.61-0.70 = poor; and 0.50-0.60 = fail.[20,21] Statistical significance was considered at an alpha level of 0.05, and all analyses were performed in STATA version 14 (StataCorp LP, College Station, Texas).

RESULTS

Video Analysis

A total of 251 videos were reviewed for VS of SRC by 2 expert raters. These videos included 211 diagnosed SRCs compared to 40 injury plays that involved player removal for evaluation of a suspected concussion but without a diagnosed SRC.

Frequency, Sensitivity, and Specificity of the VS Checklist

The frequencies and percentages of each VS recorded by rater 1 and rater 2 are presented in Table 2. An examination of VS, regardless of SRC group, revealed that slow to get up, clutching at the head/face, and no behavior observed were the most frequent VS of possible SRC. Motor incoordination was amongst the most frequently observed by both raters for concussed NFL players yet was not frequently observed (ie, recorded in less than 10% of players by both raters) in nonconcussed players. The least frequently recorded VS of SRC were facial injury, vacant look, and impact seizure. In addition, 24% to 29% of concussed NFL players did not exhibit a VS, and 33% to 38% of nonconcussed NFL players did not exhibit a VS. The sensitivity, specificity, PPV, and NPV values for each VS are presented in Tables 3 and 4. Slow to get up, motor incoordination, and clutching at head were the VS with the highest sensitivity for SRC, and impact seizure, vacant look, loss of responsiveness, and no protective action were the most specific for SRC. The VS with the highest combination of sensitivity and specificity was motor incoordination. Examining the data across both raters revealed that all VS had PPV values ranging from 50% to 100% and NPV values ranging from 14% to 21%. The overall sensitivity and specificity

TABLE 2. A Comparison of the Frequency and Percentage of the VS of SRC Recorded by 2 Expert Raters for NFL Players That Were Diagnosed (n = 211) and Not Diagnosed With SRC (n = 40) During the 2017 Season

	SRC diagnosis (n = 211)		No SRC diagnosis (n = 40)	
	Rater 1	Rater 2	Rater 1	Rater 2
No behavior	51 (24.2%)	61 (28.9%)	15 (37.5%)	13 (32.5%)
Slow to get up	145 (68.7%)	133 (63.0%)	24 (60.0%)	24 (60.0%)
Motor incoordination	71 (33.6%)	49 (23.2%)	3 (7.5%)	3 (7.5%)
Loss of responsiveness	40 (19.0%)	39 (18.5%)	2 (5.0%)	0 (0.0%)
No protective action	47 (22.3%)	27 (12.8%)	3 (7.5%)	2 (5.0%)
Clutching at head/face	55 (26.1%)	70 (33.2%)	11 (27.5%)	15 (37.5%)
Blank/vacant look	3 (1.4%)	4 (1.9%)	0 (0.0%)	0 (0.0%)
Impact seizure	12 (5.7%)	12 (5.7%)	0 (0.0%)	0 (0.0%)
Facial injury	2 (0.9%)	1 (0.5%)	1 (2.5%)	1 (2.5%)

Downloaded from https://academic.oup.com/neurosurgery/advance-article-abstract/doi/10.1093/neuros/nyaa072/5820431 by Ohio State University - OARDC user on 20 May 2020

TABLE 3. Comparison of VS of SRC Recorded by 2 Expert Raters for NFL Players That Were Diagnosed (n = 211) and Not Diagnosed With SRC (n = 40) During the 2017 Season: Summary of True Positive, False Negative, False Positive, True Negative, and Totals

	SRC with positive sign		SRC with negative sign		No SRC with positive sign		No SRC with negative sign			
	True positive		False negative		False positive		True negative		Totals	
	Rater 1	Rater 2	Rater 1	Rater 2	Rater 1	Rater 2	Rater 1	Rater 2	Rater 1	Rater 2
No behavior	51	61	160	150	15	13	25	27	251	251
Slow to get up	145	133	66	78	24	24	16	16	251	251
Motor incoordination	71	49	140	162	3	3	37	37	251	251
Loss of responsiveness	40	39	171	172	2	0	38	40	251	251
No protective action	47	27	164	184	3	2	37	38	251	251
Clutching at head/face	55	70	156	141	11	15	29	25	251	251
Blank/vacant look	3	4	208	207	0	0	40	40	251	251
Impact/seizure	12	12	199	199	0	0	40	40	251	251
Facial injury	2	1	209	210	1	1	39	39	251	251

TABLE 4. Comparison of VS of SRC Recorded by 2 Expert Raters for NFL Players That Were Diagnosed (n = 211) and Not Diagnosed With SRC (n = 40) During the 2017 Season: Summary of Sensitivity, Positive Predictive Value, Negative Predictive Value, Positive Likelihood Ratio, and Negative Likelihood Ratio

	Sensitivity		Specificity		Positive predictive value		Negative predictive value		Positive likelihood ratio		Negative likelihood ratio	
	Rater 1	Rater 2	Rater 1	Rater 2	Rater 1	Rater 2	Rater 1	Rater 2	Rater 1	Rater 2	Rater 1	Rater 2
No behavior	0.24 (0.19-0.31)	0.29 (0.23-0.36)	0.62 (0.46-0.77)	0.68 (0.51-0.81)	0.77 (0.77-0.78)	0.82 (0.82-0.83)	0.14 (0.13-0.14)	0.15 (0.15-0.15)	0.64 (0.64-0.65)	0.88 (0.88-0.90)	1.21 (1.20-1.22)	1.05 (1.05-1.06)
Slow to get up	0.69 (0.62-0.75)	0.63 (0.56-0.70)	0.40 (0.25-0.57)	0.40 (0.25-0.57)	0.86 (0.86-0.86)	0.85 (0.85-0.85)	0.20 (0.19-0.20)	0.17 (0.17-0.17)	1.15 (1.14-1.16)	1.05 (1.04-1.06)	0.78 (0.77-0.79)	0.92 (0.91-0.94)
Motor incoordination	0.34 (0.27-0.40)	0.23 (0.18-0.30)	0.92 (0.80-0.98)	0.92 (0.80-0.98)	0.96 (0.96-0.96)	0.94 (0.94-0.94)	0.21 (0.20-0.21)	0.19 (0.18-0.19)	4.48 (4.33-4.65)	3.10 (2.99-3.21)	0.71 (0.71-0.72)	0.83 (0.83-0.83)
Loss of responsiveness	0.19 (0.14-0.25)	0.18 (0.13-0.24)	0.95 (0.83-0.99)	1.00 (0.91-1.00)	0.95 (0.95-0.95)	1.00 –	0.18 (0.18-0.18)	0.19 (0.19-0.19)	3.79 (3.63-3.96)	–	0.85 (0.85-0.86)	0.82 (0.81-0.82)
No protective action	0.22 (0.17-0.28)	0.13 (0.09-0.18)	0.92 (0.80-0.98)	0.95 (0.83-0.99)	0.94 (0.94-0.94)	0.93 (0.93-0.93)	0.18 (0.18-0.19)	0.17 (0.17-0.17)	2.97 (2.87-3.08)	2.56 (2.45-2.68)	0.84 (0.84-0.84)	0.92 (0.92-0.92)
Clutching at head/face	0.26 (0.20-0.33)	0.33 (0.27-0.40)	0.72 (0.56-0.85)	0.62 (0.46-0.77)	0.83 (0.83-0.84)	0.82 (0.82-0.83)	0.16 (0.16-0.16)	0.15 (0.15-0.15)	0.95 (0.93-0.96)	0.88 (0.87-0.90)	1.02 (1.01-1.03)	1.07 (1.06-1.08)
Blank vacant look	0.01 (0.00-0.04)	0.02 (0.01-0.05)	1.00 (0.91-1.00)	1.00 (0.91-1.00)	1.00 –	1.00 –	0.16 (0.16-0.16)	0.16 (0.16-0.16)	–	–	0.98 (0.99-0.99)	0.98 (0.98-0.99)
Impact seizure	0.06 (0.03-0.10)	0.06 (0.03-0.10)	1.00 (0.91-1.00)	1.00 (0.91-1.00)	1.00 –	1.00 –	0.16 (0.17-0.17)	0.16 (0.17-0.17)	–	–	0.94 (0.94-0.94)	0.94 (0.94-0.94)
Facial injury	0.01 (0.00-0.03)	0.00 (0.00-0.03)	0.97 (0.87-1.00)	0.97 (0.87-1.00)	0.67 (0.65-0.68)	0.50 (0.50-0.50)	0.16 (0.16-0.16)	0.15 (0.15-0.16)	0.38 (0.35-0.41)	0.19 (0.17-0.21)	1.02 (1.01-1.02)	1.02 (1.02-1.02)

Downloaded from https://academic.oup.com/neurosurgery/advance-article-abstract/doi/10.1093/neuros/nyaa072/5820431 by Ohio State University - OARDC user on 20 May 2020

TABLE 5. The Sensitivity and Specificity of VS Checklist (Yes/No) for Identifying Concussion in NFL Players Who Were Diagnosed (n = 211) and Not Diagnosed With SRC (n = 40) During the 2017 Season

	Rater 1	Rater 2	Combined[a]
Sensitivity	75.83	71.09	73.46
Specificity	62.50	67.50	65.00

[a]Combined is the average of scores between raters 1 and 2.

for the VS checklist (ie, exhibiting at least one VS) for predicting SRC were 73% and 65%, respectively (see Table 5).

Documenting the Contribution of Each VS for Predicting SRC

The results from univariate and multivariate logistic regression with corresponding beta values are presented in Table 6. Multivariate analysis revealed the significant predictors of SRC included 2 VS: motor incoordination (odds ratio [OR] 3.96, 95% CI 1.64, 9.56, $P = .002$) and loss of responsiveness (OR 5.17, 95% CI 1.14, 23.42, $P = .033$). An ROC curve predicting the ability to diagnose SRC based on the 8 VS, excluding no behavior, generated an AUC of 0.66, leading to "poor" ability to predict SRC based on VS (see Figure).

DISCUSSION

The aim of the current study was to document the utility of a VS checklist for identifying SRC in NFL players. Frequency, sensitivity, specificity, and overall predictive ability were reported. According to the ratings of 2 league-appointed experts, the most frequently recorded VS in concussed athletes were slow to get up, clutching at head/face, and motor incoordination, whereas the least frequent recorded VS were facial injury, impact seizure, and blank/vacant look. The VS of slow to get up, motor incoordination, and clutching at the head were the most sensitive for SRC, whereas impact seizure, blank/vacant look, loss of responsiveness, and no protective action were the most specific for SRC. However, 26% of concussed NFL players exhibited no VS. Moreover, the sensitivity and specificity for exhibiting any VS for SRC were 76% and 63%, respectively, with an AUC of 0.66, classified as poor predictive ability.[20,21] Despite promising results, in its early and non-NFL specific form, the current VS checklist alone appears inadequate to identify SRC in NFL players.

Frequency of VS

The frequencies of the VS documented in the current study of NFL players appear comparable to other professional leagues. The most common VS were slow to get up, no behavior, clutching at head/face, and motor incoordination, suggesting agreement across other professional leagues such as the AFL[9] and NHL[10]. However, the frequency of some VS signs (eg, slow to get up) may be artificially higher in other sports because of a competitive advantage for staying on the ground.[13] The least frequently observed VS in concussed NFL athletes included facial injury, impact seizure, and blank/vacant look, which are similar to the findings in concussed AFL players.[19] The relative infrequency of these VS in NFL players is not surprising due to the use of helmets and protective facemasks/visors. In nonconcussed athletes, the most frequent VS included slow to get up and clutching at head/face, and the least frequent were impact seizure, blank vacant look, loss of responsiveness, and motor incoordination. Similar to the current study, Davis et al[19] also reported that slow to get up and clutching at head/face were frequently observed in nonconcussed AFL players.

Sensitivity/Specificity of VS

In addition to frequency, the sensitivity and specificity of the VS for SRC reported in the current study of NFL players share similarities and differences with reports from other professional leagues. Overall, no VS had both high sensitivity and high specificity, which was similar to trends reported in NHL[10] and AFL players.[8] Researchers posit that a high sensitivity for a VS is a good indicator for identifying a potential SRC, but a low specificity suggests that the VS is not useful for a definitive diagnosis.[7,13] Given the complexity and granularity of each sign and

TABLE 6. Univariate and Multivariate Logistic Regression to Test Association Between Each VS and SRC Diagnosis

	Univariate		Multivariate	
	Beta (95% CI)	*P* value	Beta (95% CI)	*P* value
Slow to get up	0.25 (−0.23, 0.74)	.313	−0.16 (−0.71, 0.39)	.559
Motor incoordination	1.59 (0.73, 2.45)	<.001	1.38 (0.49, 2.26)	.002
Loss of responsiveness	2.20 (0.77, 3.62)	.003	1.64 (0.13, 3.15)	.033
No protective action	1.16 (0.22, 2.10)	.016	0.19 (−0.86, 1.25)	.717
Clutching at head/face	−0.13 (−0.65, 0.38)	.607	−0.21 (−0.78, 0.36)	.465
Blank/vacant look	–	–	–	–
Impact seizure	–	–	–	–
Facial injury	−1.27 (−3.08, 0.53)	.166	−0.73 (−2.56, 1.10)	.435

Downloaded from https://academic.oup.com/neurosurgery/advance-article-abstract/doi/10.1093/neuros/nyaa072/5820431 by Ohio State University - OARDC user on 20 May 2020

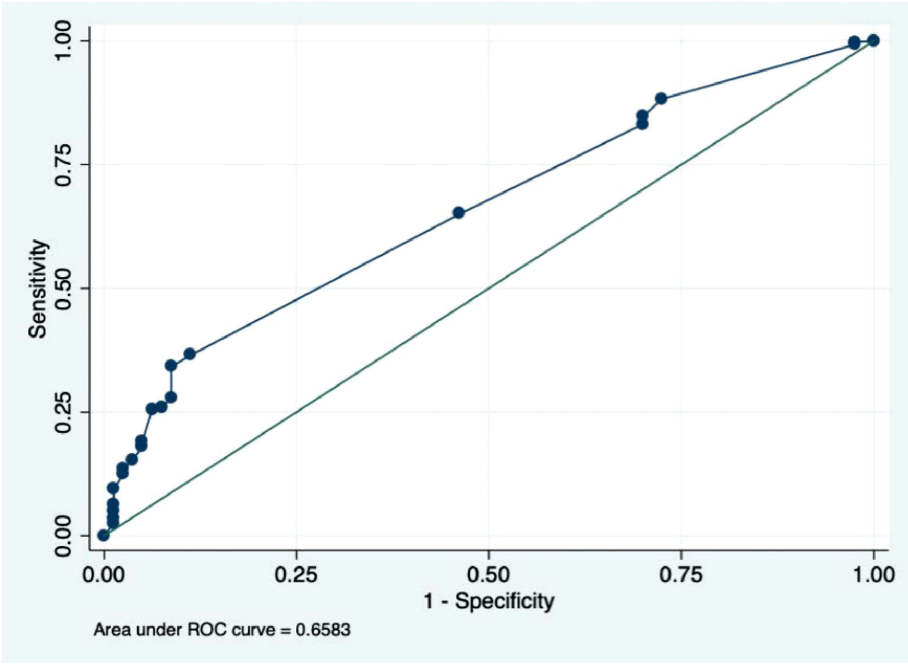

FIGURE. *Receiver operating characteristic curve (ROC) showing ability of 8 VS to predict concussion diagnosis with AUC = 0.6583.*

corresponding accuracy, we have summarized all published data from other professional leagues in Table 7.

To note some important trends, motor incoordination appeared to have the optimal combination of sensitivity and specificity, with PPV of 96% and 94%, which was corroborated by the AFL study,[7] and was one of 2 VS that remained significant on multivariate testing. Interestingly, motor incoordination had a PPV of 31% in NHL players, highlighting difference in assessing players on ice skates vs foot. We can also group loss of responsiveness and impact seizure together as being extremely specific, yet, at the cost of poor sensitivity, similar to AFL findings.[7] Blank/vacant look has potential to have high PPV, yet agreement can be problematic, as reported in previous NFL[17] and AFL studies.[7] Nonetheless, VS with high specificity should be useful in determining removal from play, especially in players with an SRC who may otherwise continue to play.

Predictive Ability of VS

In summarizing the results of our study, the use of a VS checklist alone in its current form appears inadequate to identify an SRC. A sizable percentage of SRC (26%) displayed no observable VS. Although this number is an improvement from an NHL study in which 53% of SRC exhibited no VS, it is still unacceptable to fail to identify 1 in 4 injuries solely via VS methodology.[10] Unfortunately, the percentage of SRC with no observable VS was not reported in the AFL study using 8 VS[7] and in the NRL study using 6 VS[13]; therefore, additional discussion is needed to discern what proportion of SRC may be missed with the various checklists. The reported AUC of VS to accurately predict SRC between both raters was classified as "poor."[21] This low percentage could be for several reasons. First, specific American football gameplay and equipment differences may require a different, more specific checklist, and previously published checklists may not be generalizable. Second, only 2 expert reviewers were used in the current study, and the use of more expert reviewers may yield different results. Third, the current study evaluated data from a single season, and the results could be altered by a larger sample size and the inclusion of multiple seasons. Fourth, the detailed video capture of every event that occurs among every player across all plays on an NFL field may be more aspirational than realistic. Fifth, it is also possible that several VS (eg, slow to get up and adjusting the helmet/clutching at the face) are seen after musculoskeletal or other injuries and not related to SRC. Taken together, data from the current study and others[7,10] reveal that the use of VS alone is medically inappropriate and inferior to a proper sideline or acute medical evaluation as a more objective and thorough method for determining when an athlete has sustained an SRC.

Future Research

There are several future directions for this research. Because of the timeframe of data collection for the current study,

Downloaded from https://academic.oup.com/neurosurgery/advance-article-abstract/doi/10.1093/neuros/nyaa072/5820431 by Ohio State University - OARDC user on 20 May 2020

TABLE 7. Sensitivity, Specificity, Positive Predictive Value, and Negative Predictive Value for Other Leagues

Author, year[a]	Sport	No behavior seen	Slow to get up	Motor incoordination	Loss of responsiveness	No protective action	Clutching at head/face	Blank/vacant look	Impact seizure	Facial injury
Makdissi and Davis, 2016[7,b]	AFL	–	0.87/0.19 0.46/0.65	0.29/0.95 0.81/0.63	0.40/0.75 0.56/0.61	–	0.20/0.67 0.32/0.51	0.09/1.00 1.00/0.58	0.07/0.93 0.56/0.61	0.20/0.84 0.50/0.57
Gardner et al, 2017[13,c]	NRL	–	1.00/0.50 0.27/1.00	0.58/0.79 0.34/0.91	0.40/0.91 0.46/0.89	–	0.63/0.46 0.18/0.87	0.75/0.84 0.46/0.95	0.05/1.00 0.75/0.85	–
Bruce et al, 2018[11,d]	NHL	–	–	0.37/0.90 0.31/0.92	0.14/0.98 0.52/0.90	–	–	–	–	–
Echemendia et al, 2018[10,d]	NHL	53%	0.40/0.07 0.10/0.32	0.17/0.90 0.31/0.81	0.06/0.98 0.52/0.80	–	0.08/0.86 0.13/0.78	0.01/1.00 0.67/0.79	–	0.05/0.80 0.06/0.76
Current study[e]	NFL	26%	0.66/0.40 0.86/0.19	0.29/0.92 0.95/0.20	0.19/0.98 0.98/0.19	0.18/0.94 0.94/0.18	0.30/0.67 0.83/0.16	0.02/1.00 1.00/0.16	0.06/1.00 1.00/0.16	0.01/0.97 0.59/0.16

[a]Boxes organized as sens/spec (top) and PPV/NPV (below); AFL = Australian Football League; NRL = National Rugby League; NHL = National Hockey League.
[b]Makdissi and Davis[7] evaluated rag doll appearance with the following data: 0.16/0.91; 0.58/0.58.
[c]Gardner et al[13]: Motor incoordination was referred to as gait ataxia; loss of responsiveness was referred to as unresponsiveness.
[d]Echemendia et al[10] and Bruce et al[11]: Loss of responsiveness was referred to as loss of consciousness/motionless; motor incoordination was referred to as incoordination/balance problems.
[e]Represents the average of raters 1 and 2.

a group of 13 expert representatives from 7 international sporting bodies reached consensus (ie, >90% agreement) on the inclusion and definition of 6 VS of SRC that included as follows: lying motionless, motor incoordination, impact seizure/convulsion, tonic posturing, no protective action (floppy and tonic), and blank/vacant look.[22] Although there is significant overlap between this recent list of VS[22] and the ones used in the current study, the VS definitions may differ (eg, the amount of time lying motionless). Future research examining the VS in NFL players should incorporate these VS[22] and their definitions to standardize VS evaluation across all professional sports. Previous NHL researchers report that the inclusion of mechanism of injury (MOI) in addition to the VS was a superior approach to establishing a risk-prediction model,[11,23,24] which should be investigated going forward in NFL players. Future studies should also include an analysis of multiple VS, which is similar to Echemendia et al[10] and conduct the investigation across multiple seasons.

Limitations

There are several limitations to the current study. First, there were 2 expert raters who identified VS of SRC, and this study did not include a third rater to achieve consensus when there was inter-rater disagreement. Second, the results of this study were limited to data from a single NFL season, and only included players suspected of having an SRC and who were removed for medical evaluation. All head impacts from the entire NFL season, irrespective of magnitude, were not reviewed. Finally, the results of a NFL-based study may not be generalizable to other levels of American football play.

CONCLUSION

These data are among the first attempt to examine the sensitivity and specificity of the AFL VS checklist in NFL players and indicate that identifying a SRC solely via a VS checklist conducted through video review is often insufficient for diagnosis. The assessment of VS (or lack thereof) is only one part of the sideline/acute evaluation of SRC in NFL players. Moreover, the lack of VS observed in 26% of concussed NFL players emphasizes the importance of the sideline evaluation for SRC in the NFL. Therefore, revision or creation of a VS checklist is needed to better identify the features of concussive injury that may or may not be visible to sports medicine and video review personnel (ie, facial injury in a helmeted sport). There is a need to continue to improve head injury safety in all levels of American football, and the examination of the current VS of SRC in NFL players is an ideal place to start. As our understanding of concussion pathophysiology improves, it may be possible to identify other VS with higher sensitivity and specificity to SRC.

Disclosures

Dr Sills is the Chief Medical Officer and a full time employee of the National Football League. Drs Crandall and Lessley are affiliated with Biomechanics Consulting and Research, which has a consulting contract with the National Football League. Dr Solomon is a paid consultant to the National Football League Health and Safety Department. The other authors have no personal, financial, or institutional interest in any of the drugs, materials, or devices described in this article.

REFERENCES

1. Elbin RJ, Sufrinko A, Schatz P, et al. Removal from play after concussion and recovery time. *Pediatrics*. 2016;138(3):e20160910.

Downloaded from https://academic.oup.com/neurosurgery/advance-article-abstract/doi/10.1093/neuros/nyaa072/5820431 by Ohio State University - OARDC user on 20 May 2020

2. Asken BM, Bauer RM, Guskiewicz KM, et al. Immediate removal from activity after sport-related concussion is associated with shorter clinical recovery and less severe symptoms in collegiate student-athletes. *Am J Sports Med*. 2018;46(6):1465-1474.
3. McCrea M, Hammeke T, Olsen G, Leo P, Guskiewicz K. Unreported concussion in high school football players. *Clin J Sport Med*. 2004;14(1):13-17.
4. Kerr ZY, Register-Mihalik JK, Kay MC, DeFreese JD, Marshall SW, Guskiewicz KM. Concussion nondisclosure during professional career among a cohort of former National Football League athletes. *Am J Sports Med*. 2018;46(1):22-29.
5. Kerr ZY, Register-Mihalik JK, Kroshus E, Baugh CM, Marshall SW. Motivations associated with nondisclosure of self-reported concussions in former collegiate athletes. *Am J Sports Med*. 2016;44(1):220-225.
6. Kerr ZY, Register-Mihalik JK, Marshall SW, Evenson KR, Mihalik JP, Guskiewicz KM. Disclosure and non-disclosure of concussion and concussion symptoms in athletes: review and application of the socio-ecological framework. *Brain Inj*. 2014;28(8):1009-1021.
7. Makdissi M, Davis G. The reliability and validity of video analysis for the assessment of the clinical signs of concussion in Australian football. *J Sci Med Sport*. 2016;19(10):859-863.
8. Makdissi M, Davis G. Using video analysis for concussion surveillance in Australian football. *J Sci Med Sport*. 2016;19(12):958-963.
9. Davis GA, Makdissi M, Bloomfield P, et al. International study of video review of concussion in professional sports. *Br J Sports Med*. 2019;53(20):1299-1304.
10. Echemendia RJ, Bruce JM, Meeuwisse W, Hutchison MG, Comper P, Aubry M. Can visible signs predict concussion diagnosis in the National Hockey League? *Br J Sports Med*. 2018;52(17):1149-1154.
11. Bruce JM, Echemendia RJ, Meeuwisse W, Hutchison MG, Aubry M, Comper P. Development of a risk prediction model among professional hockey players with visible signs of concussion. *Br J Sports Med*. 2018;52(17):1143-1148.
12. Gardner AJ, Howell DR, Iverson GL. A video review of multiple concussion signs in National Rugby League match play. *Sports Med - Open*. 2018;4(1):5.
13. Gardner AJ, Howell DR, Levi CR, Iverson GL. Evidence of concussion signs in national rugby league match play: a video review and validation study. *Sports Med—Open*. 2017;3(1):29.
14. Gardner AJ, Kohler R, McDonald W, Fuller GW, Tucker R, Makdissi M. The use of sideline video review to facilitate management decisions following head trauma in super rugby. *Sports Med - Open*. 2018;4(1):20.
15. Lessley DJ, Kent RW, Funk JR, et al. Video analysis of reported concussion events in the National Football League during the 2015-2016 and 2016-2017 seasons. *Am J Sports Med*. 2018;46(14):3502-3510.
16. Ellenbogen RG, Batjer H, Cardenas J, et al. National football league head, neck and spine committee's concussion diagnosis and management protocol: 2017–18 season. *Br J Sports Med*. 2018;52(14):894-902.
17. Zuckerman SL, Elbin RJ, Sills AK, et al. Concussions in the National Football League: the evolution of video review for assessing the frequency and reliability of visible signs. *Phys Sportsmed*. 2020; 1-6 (doi:10.1080/00913847.2020.1731379).
18. NFL. Head, Neck, and Spine Committee. NFL Head, Neck and Spine Committee's Protocols Regarding Diagnosis and Management of Concussion. 2013; http://static.nfl.com/static/content/public/photo/2013/10/01/0ap2000000254002.pdf. Last accessed January 10, 2020.
19. Davis G, Makdissi M. Use of video to facilitate sideline concussion diagnosis and management decision-making. *J Sci Med Sport*. 2016;19(11):898-902.
20. Mandrekar JN. Receiver operating characteristic curve in diagnostic test assessment. *J Thorac Oncol*. 2010;5(9):1315-1316.
21. Tape TG. *The Area Under an ROC Curve*. 2019; http://gim.unmc.edu/dxtests/roc3.htm. Accessed July 31, 2019.
22. Davis GA, Makdissi M, Bloomfield P, et al. Definitions of video signs of concussion in professional sports. *Br J Sports Med*. 2019;53(20):1264-1267.
23. Hutchison MG, Comper P, Meeuwisse WH, Echemendia RJ. An observational method to code concussions in the National Hockey League (NHL): the heads-up checklist. *Br J Sports Med*. 2014;48(2):125-129.
24. Hutchison MG, Comper P, Meeuwisse WH, Echemendia RJ. A systematic video analysis of National Hockey League (NHL) concussions, part I: who, when, where and what? *Br J Sports Med*. 2015;49(8):547-551.

Acknowledgments

The authors extend appreciation to Tim McMurry, PhD, for statistical expertise and to the NFL Players Association for their cooperation and assistance with the completion of this study.

Downloaded from https://academic.oup.com/neurosurgery/advance-article-abstract/doi/10.1093/neuros/nyaa072/5820431 by Ohio State University - OARDC user on 20 May 2020